Musical 'Lick' Modifier

Max Su
Advised by Scott Petersen, Dept. of Computer Science, Yale University

Introduction

Improvisation is at the heart of jazz music, yet many aspiring jazz musicians find themselves encountering creative barriers when developing their improvisational skills.

The key to overcoming these barriers often lies in mastering the use of "licks" — short, practiced phrases used repeatedly during improvisations. Licks form the foundational vocabulary for jazz musicians, serving as reliable motifs or musical ideas that can be seamlessly integrated into solos.

This project proposes that a computerized lick modifier can be an effective tool for a musician to gain inspiration.

Key Insights

> The use of Markov chains to stochastically regenerate a lick is an efficient method of computer music generation.

> User input is the only data driving the model – no external training data.

> Rule-based symbolic music generation promotes musical transparency and pedagogical value.

Background

Bebop is a style of jazz that emerged in the early 1940s as a departure from the more dance-oriented swing era. Bebop musicians emphasize essential in the chord to create solos that remain closely tied to the underlying harmony.

Because of the emphasis on ease of use and transparency, we moved forward with a rule-based system instead of a self-learning system, which would require more input data and turn into a 'black box.'

We also moved forward with a symbolic approach to organizing information rather than an audial approach. By representing music as discrete note events, we can more reliably manipulate and analyze our licks.

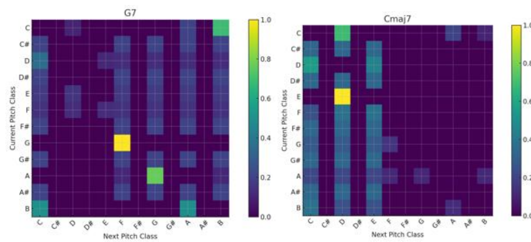
Methodology

We constructed a Python class for each step of the process. Apart from input and output, we have three steps: parsing, regeneration, and embellishing.

The parser is responsible for gathering information from the original lick and organizing it in the form of transition matrices.

The creator class is responsible for taking the transition matrices and outputting a sequence of tuples: (pitch class, note length).

The embellisher is responsible for making sure that the music is interesting. This includes adding triplets, turns, and reassigning octaves to the pitch class.



Heatmaps of the pitch class transition matrices generated by our parser for bars 2 and 3 of the example in the top right.

Results

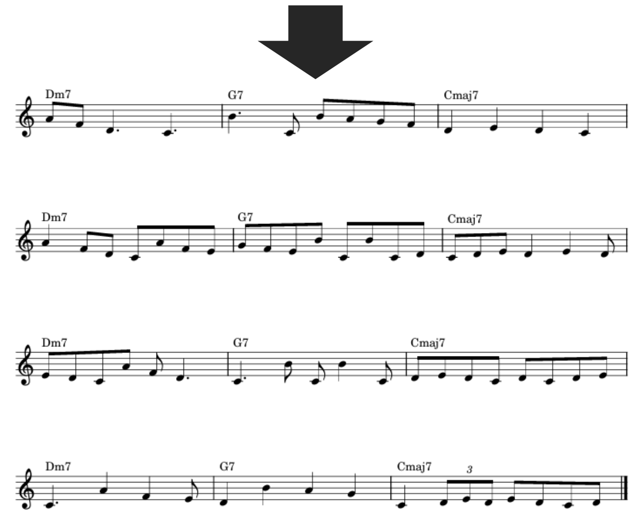
The program is able to produce numerous ideas that stylistically adhere to the convention of Bebop in seconds.

Regardless of length or chord, the program is flexible enough to adapt to what is input harmonically and melodically.

In the examples to the right, similarities exist between the original lick and the examples. For instance, in the second bar, the third of the G7 chord (B) appears and is emphasized.

The lines are also unique and can inspire new improvisations: Syncopation occurs in different places and where the chord tones lie also varies in the output licks.

Original lick input into the modifier



First 4 outputs

Conclusions and Future Directions

The results clearly demonstrate that our Markov model successfully produced musically coherent and altered jazz improvisations that carries aspects of the original lick provided.

This underscores the efficacy of leveraging probabilistic models with limited data, proving that a simple model only using data input by a musician can assist in creating new improvisations.

In choosing to use one-line MIDI data as our input, we lose the ability to read in other types of data, such as text.

Because there is no direct link between the chord changes and the music, the user can currently only have one chord per bar. More work needs to be done to refine the model and improve flexibility.